

ECE 253 – Digital Image Processing

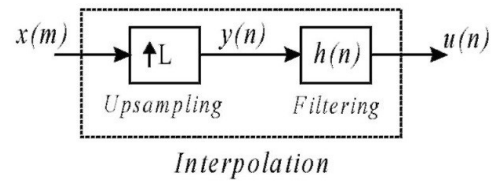
Courtesy of Pam Cosman & Ben Ochoa

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Outline

- Review of Interpolation & Decimation
- Visual Masking, Weber Law
- Intensity Transformation
- Histogram Equalization

Interpolation



Upsampling x2



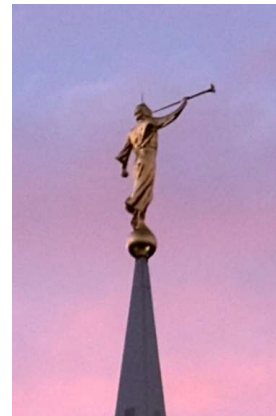
V (input)



U ($\uparrow 2$)



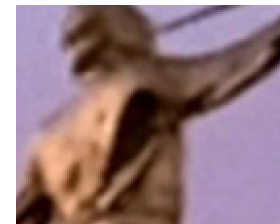
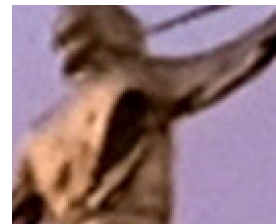
Y1 (nearest)



Y2 (bilinear)



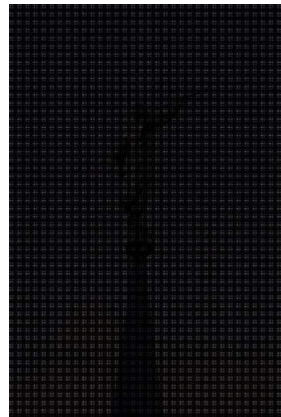
Y3 (bicubic)



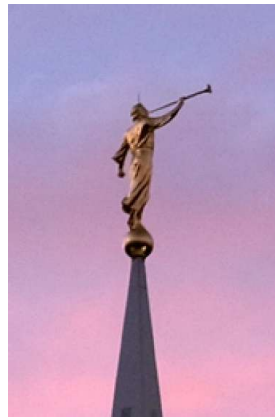
Upsampling x4



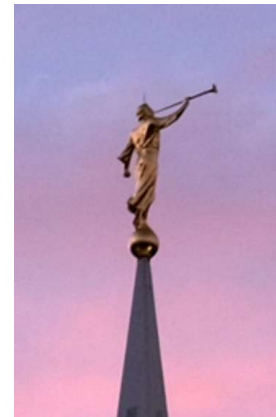
V (input)



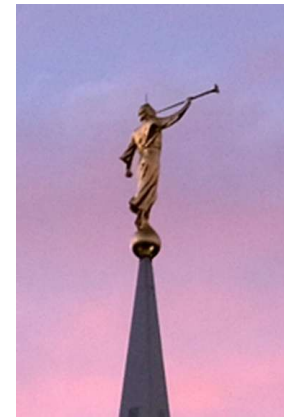
U (↑4)



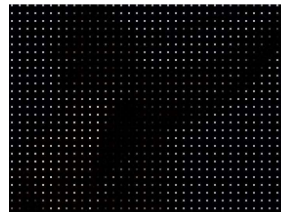
Y1 (nearest)



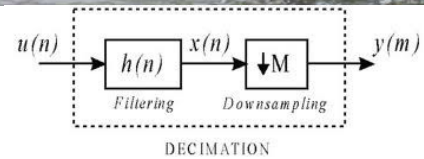
Y2 (bilinear)



Y3 (bicubic)



Decimation



Downsampling x2



X (input)



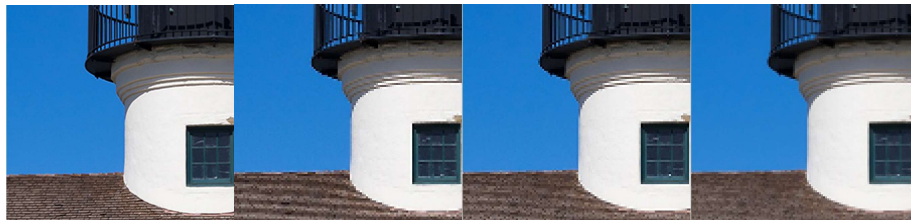
X_d (1/2)



Y₁ (bicubic)



Y₂ (AA+bicubic)



Downsampling x4



X (input)



X_d (1/4)



Y1 (bicubic)



Y2 (AA+bicubic)



Audio Rate Converter



Product Specifications

Input sample rates: between 32 kHz and 96 kHz

Output sample rates: between 32 kHz and 96 kHz (externally referenced) and either 32 kHz, 44.1 kHz, or 48 kHz, 88.2 kHz, or 96 kHz (internally referenced)

Input and output connectors: RCA,XLR, and Optical (TOSLINK)

Sync input and output connectors: XLR AES/EBU

Input and output impedance: 75 ohms (coaxial) and 110 ohms (XLR)

Input resolution: up to 24 bits

Output resolution: 24 bits

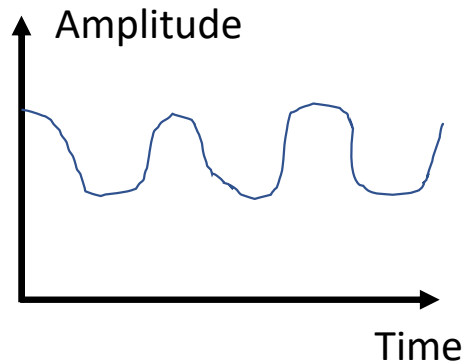
Signal-to-noise ratio: 135 dB

Total harmonic distortion + noise: -120 dB

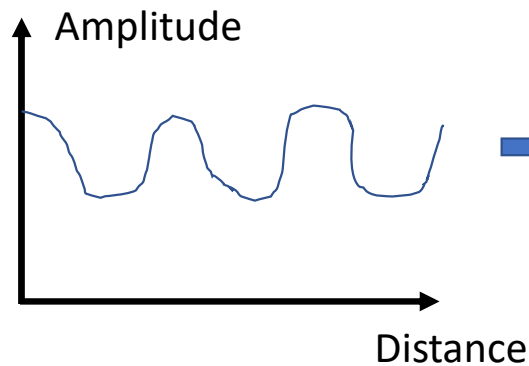
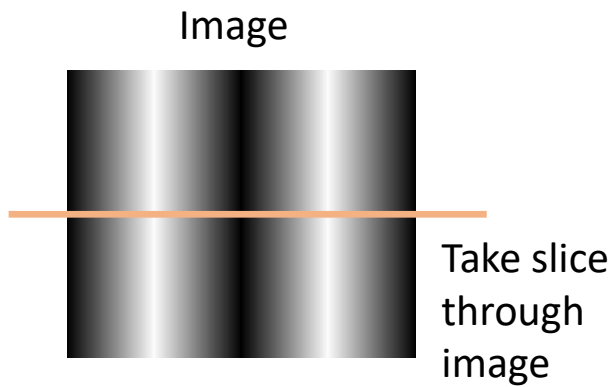
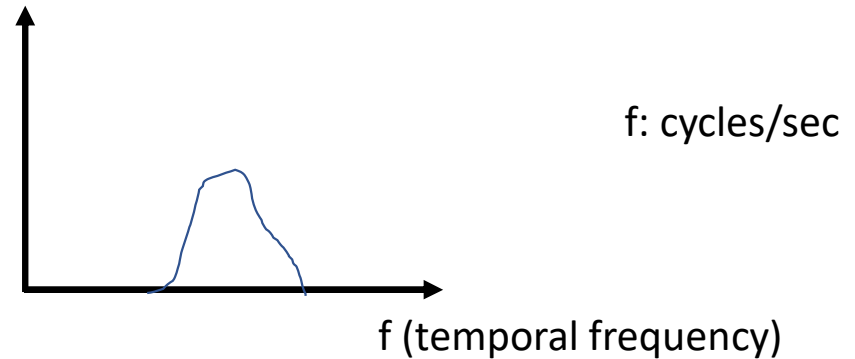
Outline

- Review of Interpolation & Decimation
- Visual Masking, Weber Law
- Intensity Transformation
- Histogram Equalization

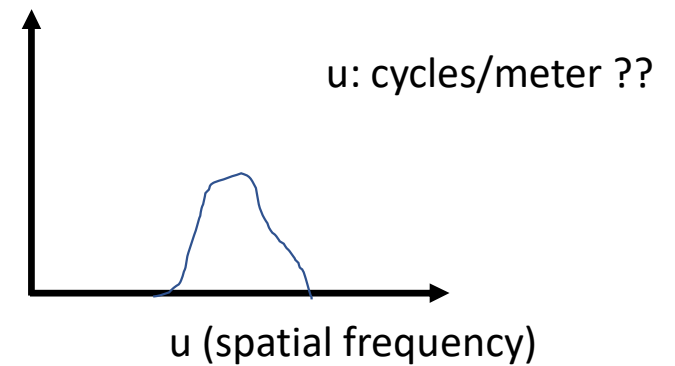
Spatial Frequency



FT
→

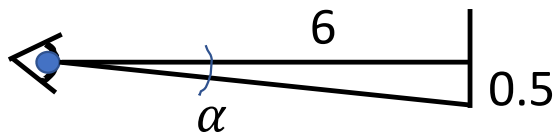


FT
→



Units for Spatial Frequency

- Analogous to temporal frequency, one might think that spatial frequency is measured in cycles per unit distance
- But we're concerned how it looks to the eye:

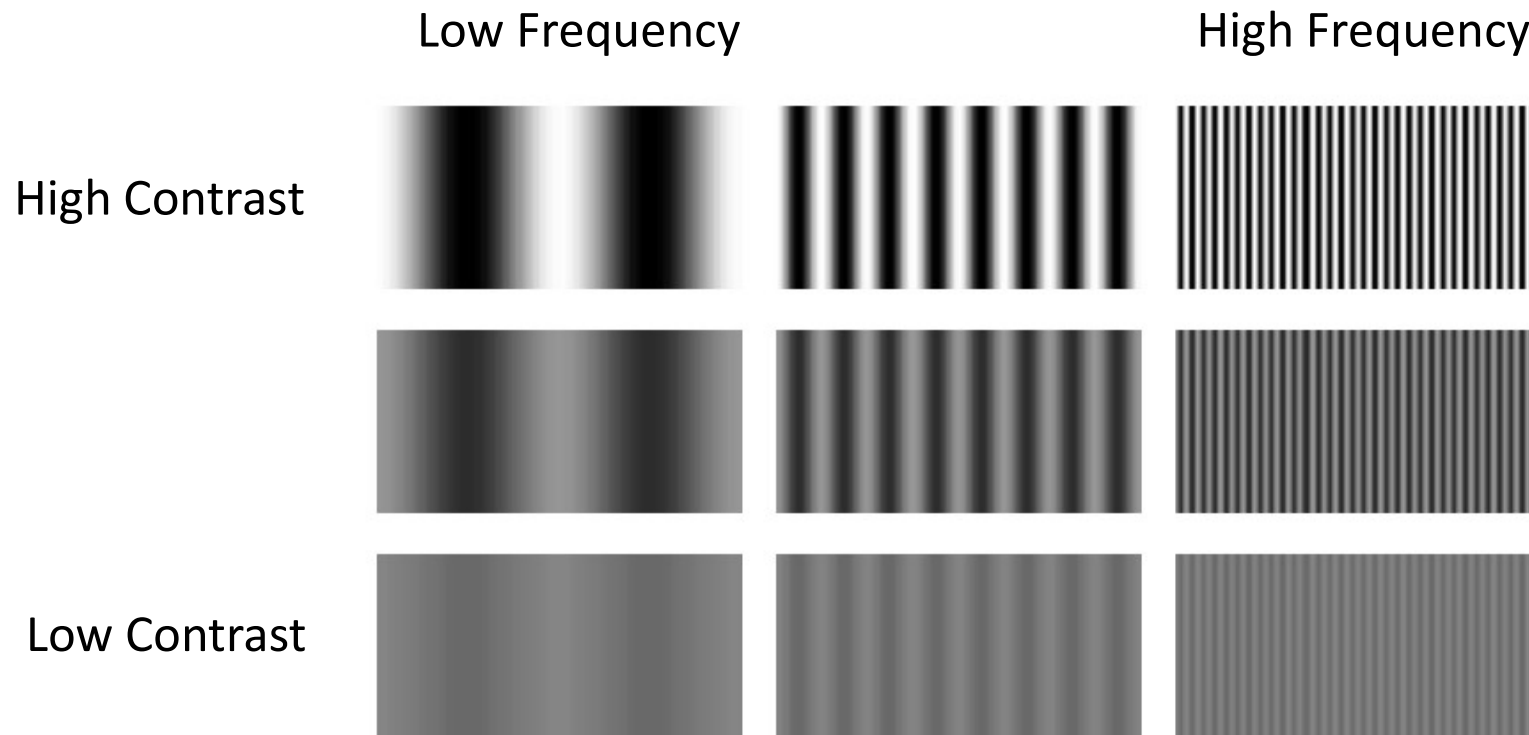


- $\tan \alpha = 0.5/6 = 1/12$
- $\alpha \approx 4.76$ degrees

- Suppose half screen width = 3600 pixels
- Suppose some pattern has 1 cycle per 320 pixels

$$\frac{3600 \text{ pixels}}{4.76 \text{ degrees}} \times \frac{1 \text{ cycle}}{320 \text{ pixels}} = 2.36 \text{ cycles per degree of visual angle} = 2.36 \text{ cpd}$$

Sensitivity to Spatial Frequency





Low
frequency

High
frequency

Credit:
Digital
Photography
School



Low
frequency

High
frequency

Credit:
Tim Burgess



279KB – 157KB
(-46%)

Quality: 80

Credit:
Tim Burgess



Low
frequency

High
frequency

Credit:
Tim Burgess



279KB – 84KB
(-70%)

Quality: 50

Credit:
Tim Burgess



Low
frequency

High
frequency

Credit:
Tim Burgess



279KB – 39KB
(-86%)

Quality: 20

Credit:
Tim Burgess

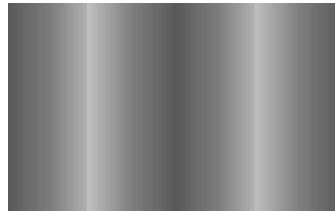
Measuring Frequency Sensitivity

Experimenter picks a spatial frequency u_0 (cycles per degree)

Observer is presented with two patterns in random order



1) Uniform gray level A



2) Sinusoidal pattern with average level A

$$L(x) = A(1 + c \sin(2\pi u_0 x))$$

A = average luminance

c = contrast ($0 < c < 1$)

$$L_{\max} = A(1 + c)$$

$$L_{\min} = A(1 - c)$$

$$c = (L_{\max} - L_{\min}) / (L_{\max} + L_{\min})$$

Measuring Frequency Sensitivity

Subject asked to say which one came first

$$L(x) = A(1 + c \sin(2\pi u_0 x))$$

Value of c is varied

$c \rightarrow 0$, subject has no idea of answer, gets 50% right

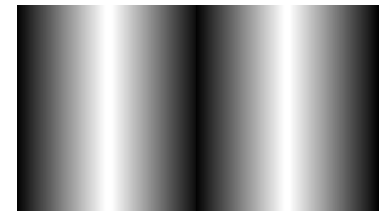
$c \rightarrow \text{Large}$, subject gets 100% right

Two alternative forced choice experiment

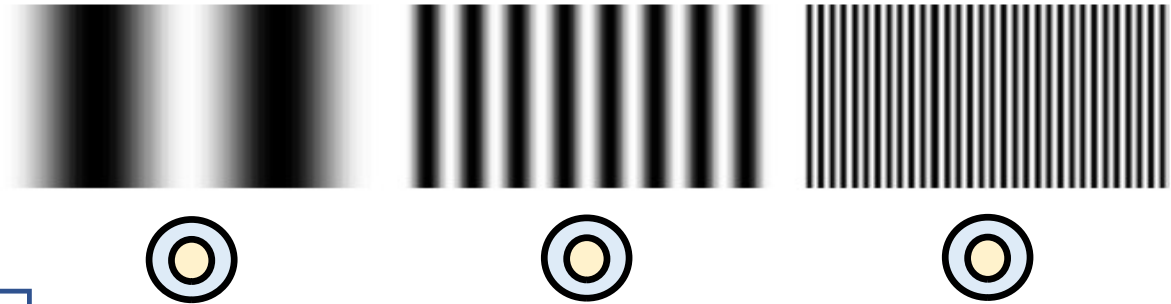
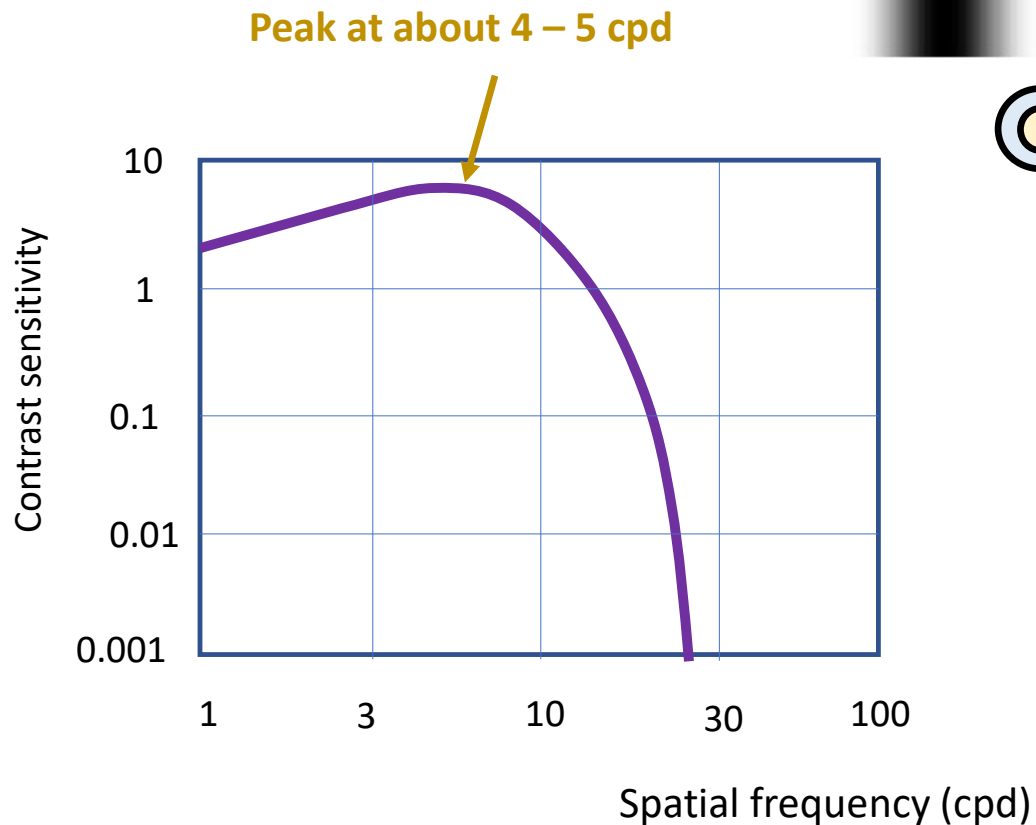
Pick a contrast c so you get target value, say 75% correct

Called contrast threshold

Reciprocal is called contrast sensitivity



Contrast Sensitivity Function



Implications for image processing?

- *Compression*: Can allow more error in high-frequency areas of an image
- *Noise reduction*: It may be hard to clean up noise in busy parts of an image. But maybe you don't need to, noise is less visible there anyway



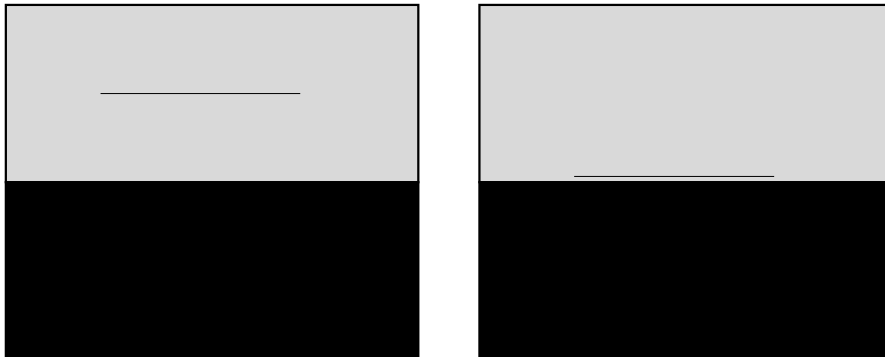
Low
frequency

High
frequency

Credit:
Giancarlo
Corti,
Unsplash

Visual Masking: Spatial Masking and Temporal Masking

- A stimulus is harder to see in the presence of large spatial changes in luminance
- Example: Line presented near a luminance edge is harder to see as it gets closer
- A stimulus is harder to see in the presence of large temporal changes in luminance
- Example: Scene cuts



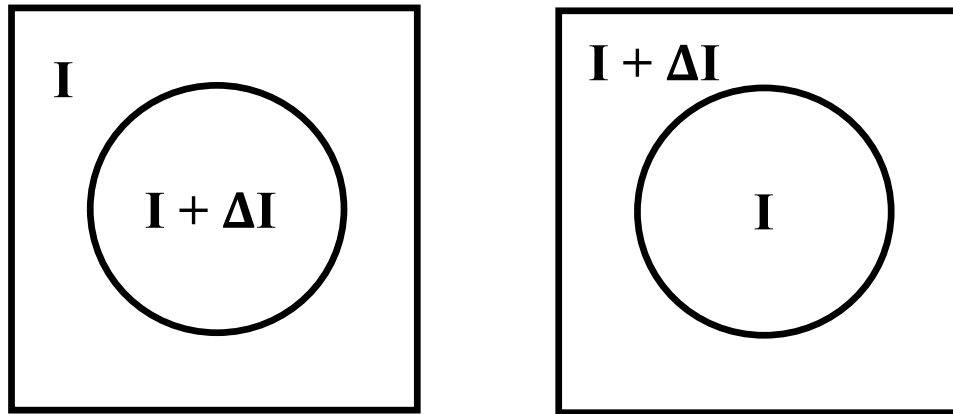
Implications for image processing?

- Noise near a high contrast edge
- Degree of compression after a scene cut, packet prioritization

Weber's Law

Experimenter chooses an intensity I

Observer is presented with one of two patterns



Increment
sensitivity

Observer is asked “Is background or circle brighter?”

Keep I fixed, try different values of ΔI

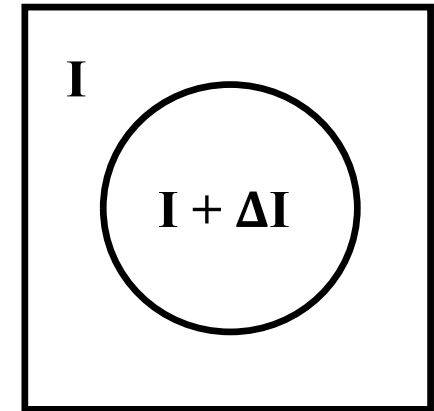
Weber's Law

Another two alternative forced choice experiment

Value of ΔI is varied

$\Delta I \rightarrow 0$, subject guesses randomly, 50% right

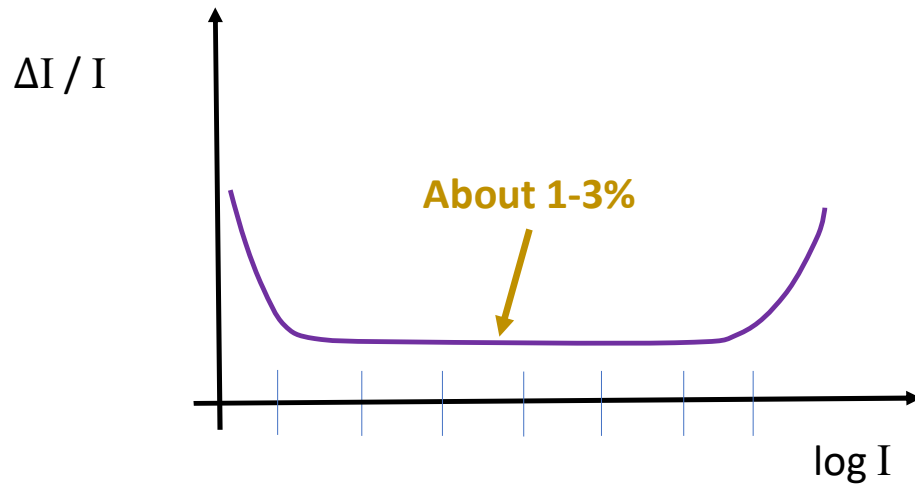
$\Delta I \rightarrow \text{Large}$, subject gets 100% right



Find ΔI so you get target value, say 75% correct

For the intensity I , this value of ΔI is called the just-noticeable difference (jnd)

Weber's Law



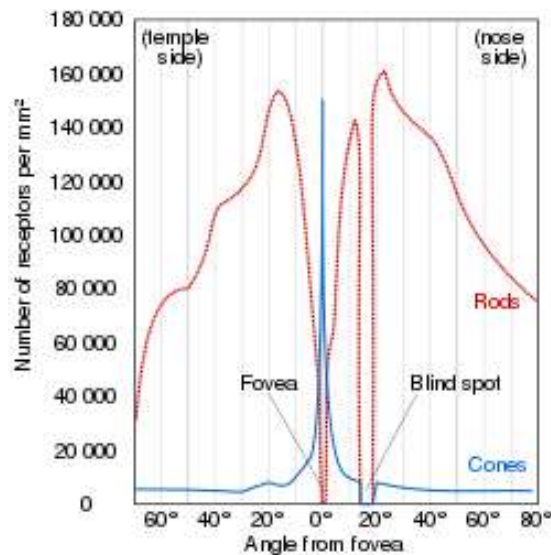
$$\frac{\Delta I}{I} \approx \text{constant} \approx 0.02$$

Implications for image processing?

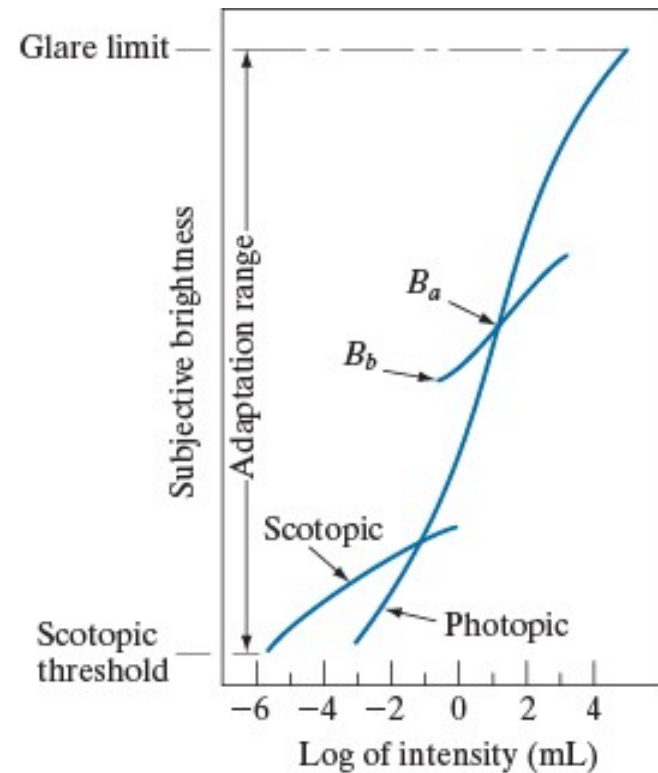
- Small differences in intensity are more noticeable in a dark region than in a bright region
- Image compression: Can allow more error in bright areas

Rods and Cones – Light Adaptation

- Retina has two main types of photoreceptors: rods (dim light) and cones (bright light)
- Overall adaptation level affects ability to detect intensity increments

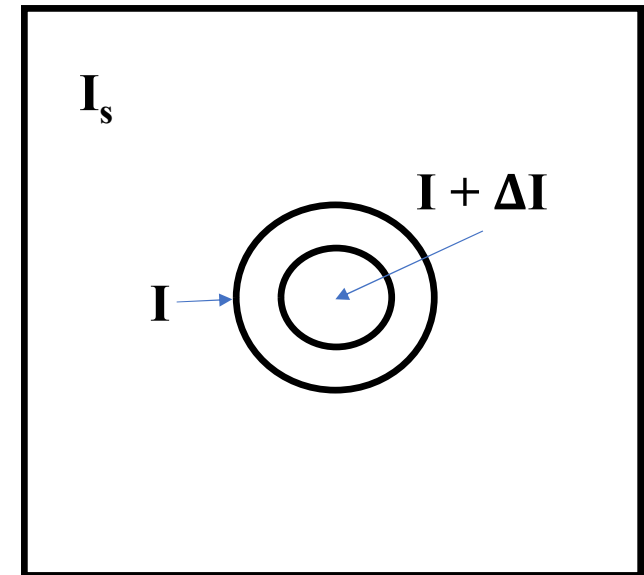
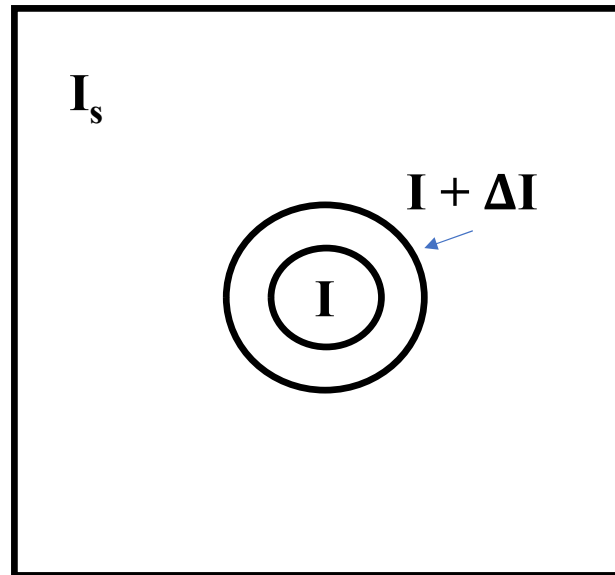


Credit:
Wikipedia,
Cmglee

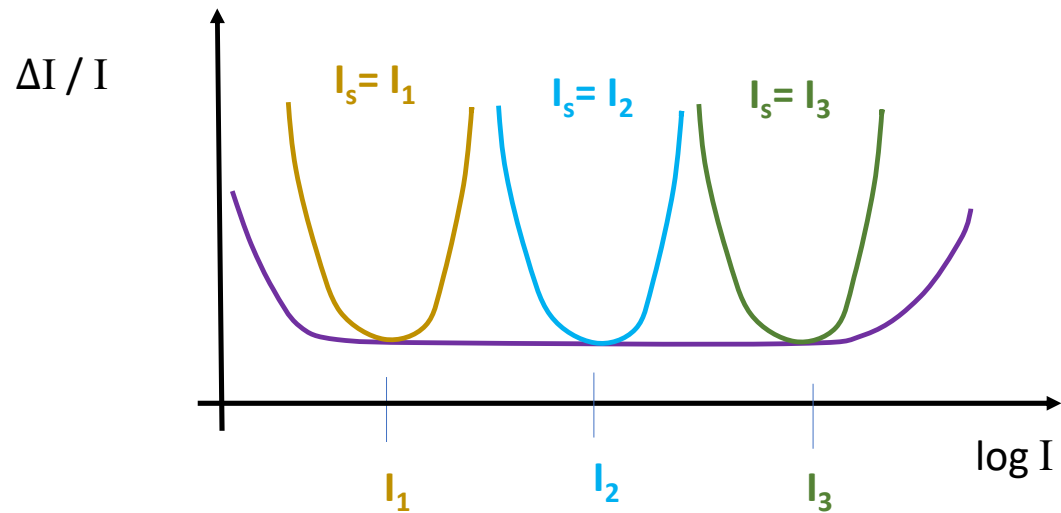


Weber's Law with Surrounds

- Large surround with intensity I_s determines adaptation level
- Observer tries to determine which circle is brighter
- Keep I and I_s fixed, try different values of ΔI



Weber's Law with Surrounds



When $I = I_1$ and $I_s = I_1$

- It's the same experiment as before

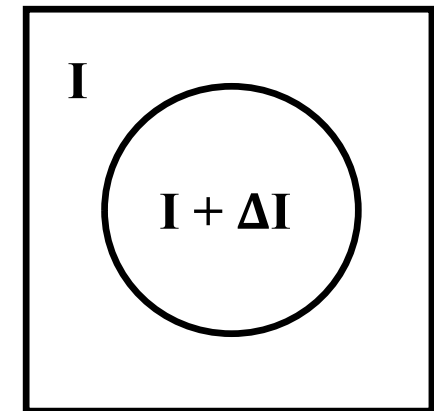
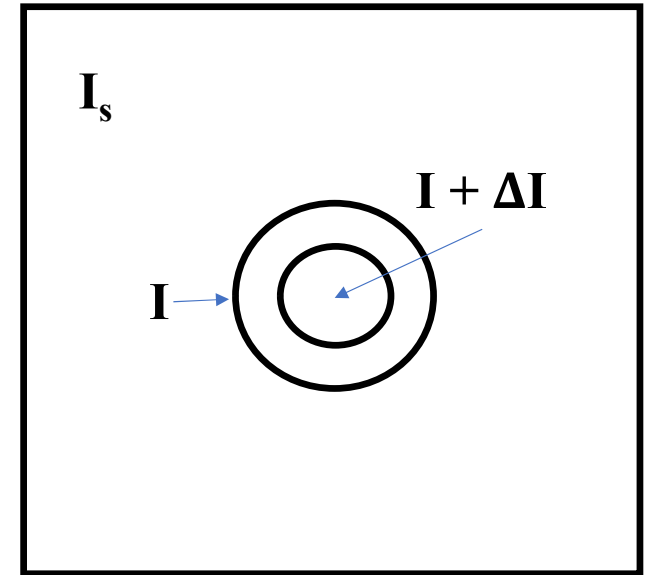


Image displayed in the dark, and in bright ambient light



Credit: Q. Song and P. C. Cosman, "Luminance Enhancement and Detail Preservation of Images and Videos Adapted to Ambient Illumination," in *IEEE Transactions on Image Processing*, vol. 27, no. 10, pp. 4901-4915, Oct. 2018

Weber's Law with Surrounds: Implications

Implications for image processing?

- Need to consider ambient lighting when designing or evaluating monitor displays or image processing algorithms
- Algorithms for the movie industry (people are dark adapted) will be different from algorithms for cell phones
- Can modify contrast values depending on ambient light

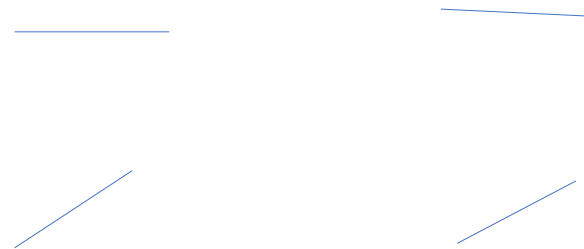
Oblique Effect

Small but consistent superiority in performance when visual stimuli are horizontal or vertical, as opposed to oblique

When a stimulus is presented at an oblique angle, people have:

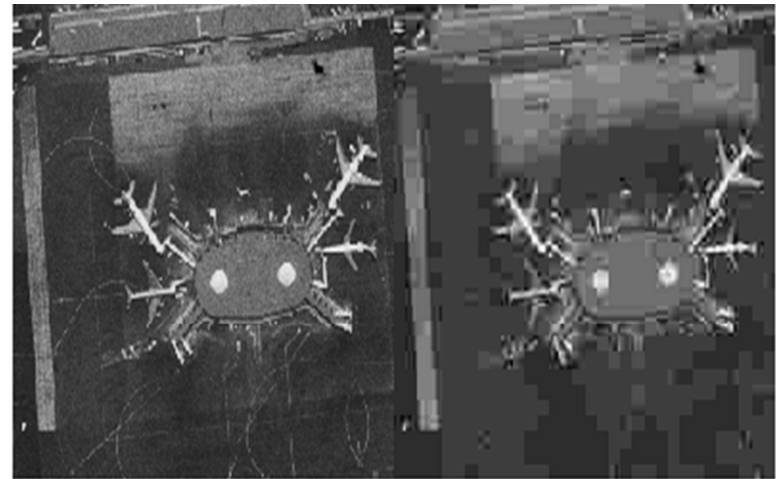
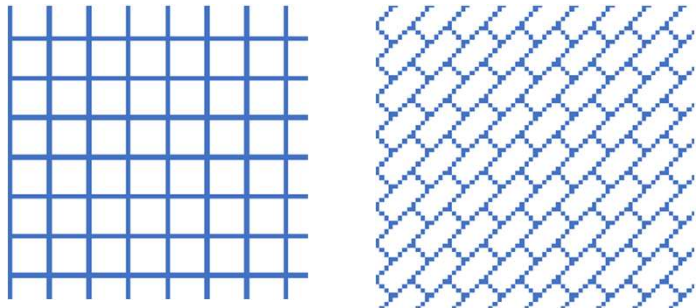
- Lowered acuity
- Reduced contrast sensitivity
- Reduced increment sensitivity

Evolutionary advantage?



Oblique Effect: Implications

Block based processing might be better using blocks with diagonal boundaries



JPEG compression example

Summary

Name	Effect
Mach bands	Enhancement of edges (overshoots and undershoots)
Frequency sensitivity	Highest sensitivity at 4-5 cpd, rapid falloff at high frequencies
Spatial masking	Reduced sensitivity to stimuli near large spatial luminance changes
Temporal masking	Reduced sensitivity to stimuli near large temporal luminance changes
Weber's Law (increment sensitivity)	just noticeable difference (jnd) is roughly proportional to background over orders of magnitude
Weber's Law with surrounds (Ambient)	larger jnd increment needed over background when you're adapted to a different ambient level
Oblique effect	greater sensitivity to stimuli oriented vertical or horizontal compared to diagonals

Outline

- Review of Interpolation & Decimation
- Visual Masking, Weber Law
- **Intensity Transformation**
- Histogram Equalization

Grayscale image processing

- Contrast enhancement
- Noise reduction
- Edge sharpening
- Mostly talking about 8-bit grayscale
- Medical images are often 10-12-bit grayscale

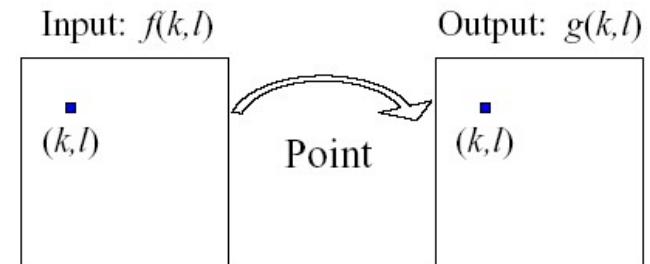
	→	White = 255 = 11111111
	→	Very light gray = 216 = 11011000
	→	Light gray = 180 = 10110100
	→	Medium light gray = 144 = 10010000
	→	Medium dark gray = 108 = 01101100
	→	Dark gray = 72 = 01001000
	→	Very dark gray = 36 = 00100100
	→	Black = 0 = 00000000

Many definitions of contrast

Definition	Notes
$c = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$	$I_{\max}$ = pattern max, $I_{\min}$ = pattern min (for sinusoidal gratings, lasers)
$(B - Ob) / B$	B = background luminance, Ob = Object luminance (for objects darker than the surround)
$B2 / B1$	any two regions, $B2 > B1$ (contrast ratio, TV engineering)
$(B1 - B2) / B_{\text{avg}}$	
$(L1 + L2) / L2$	L1 = luminous flux from lit LED, L2 = luminous flux of background reflected from LED surface
$(\text{segment} - \text{background}) / \text{segment}$	called luminance contrast

Intensity Manipulations/Transform

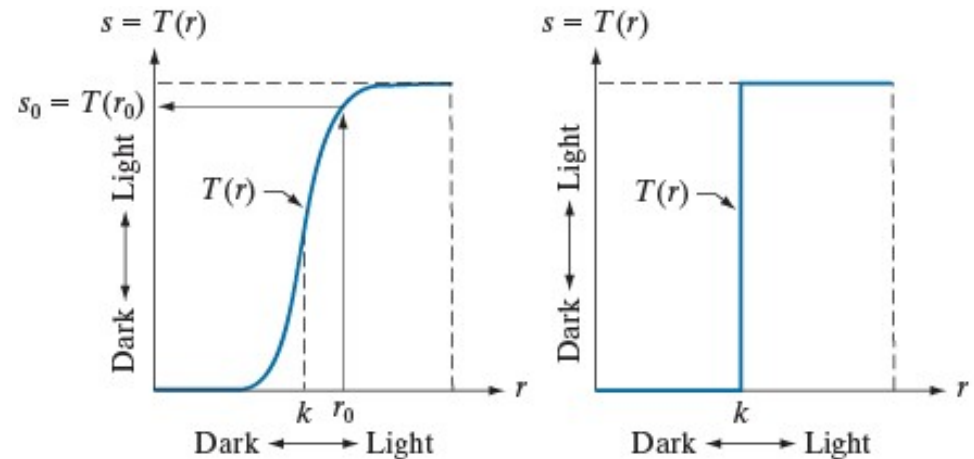
Let $f(r)$ = input gray level
 $g(s)$ = output gray level



Common to manipulate intensities using some function:

$$s = T(r)$$

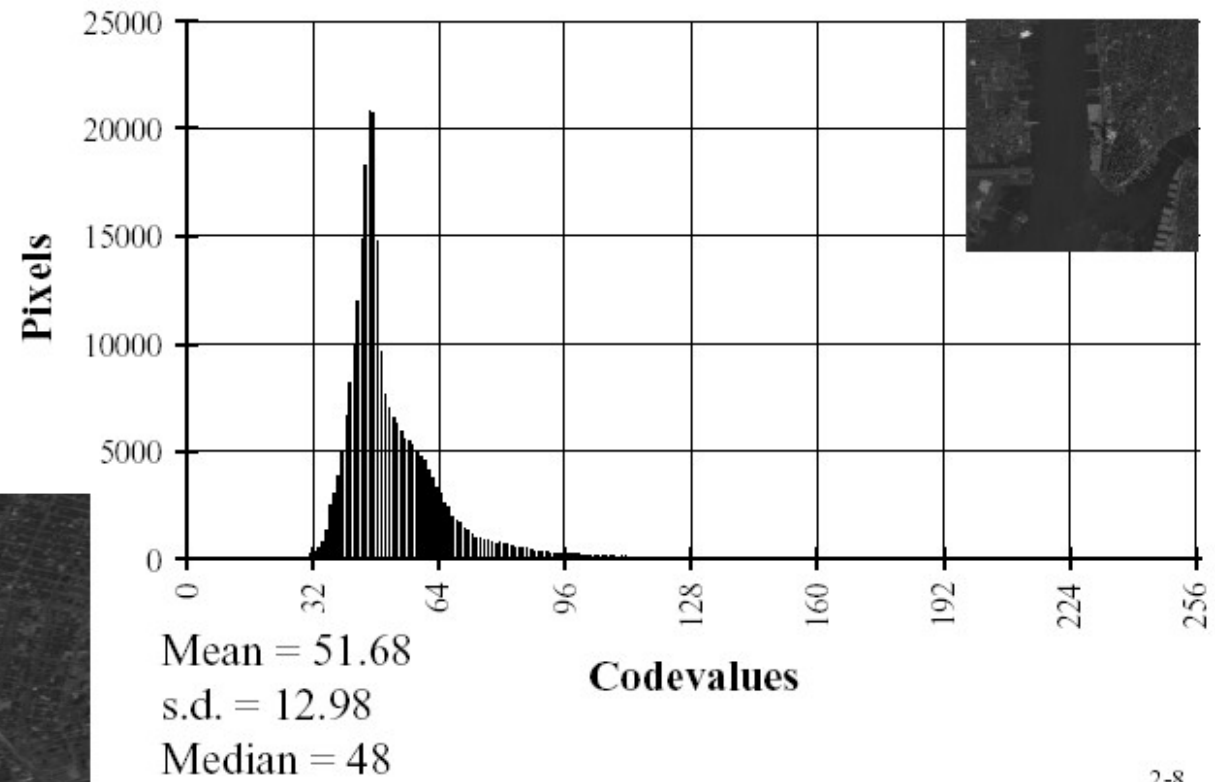
where the pixel output value does not depend on the neighborhood of the pixel but only on its intensity



Contrast stretching
function

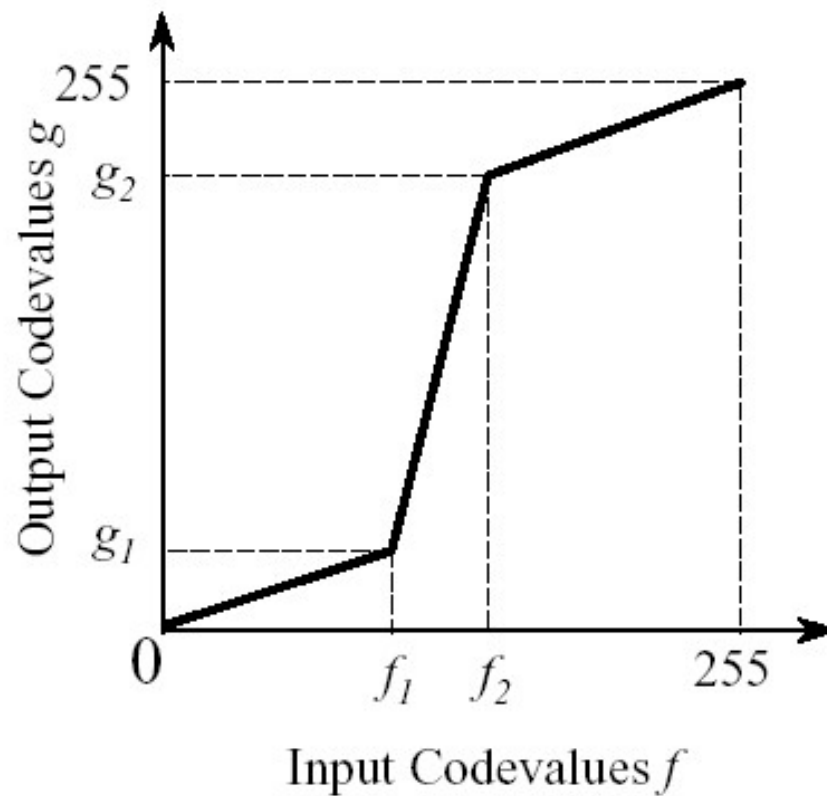
Thresholding
function

Aerial Image and its histogram



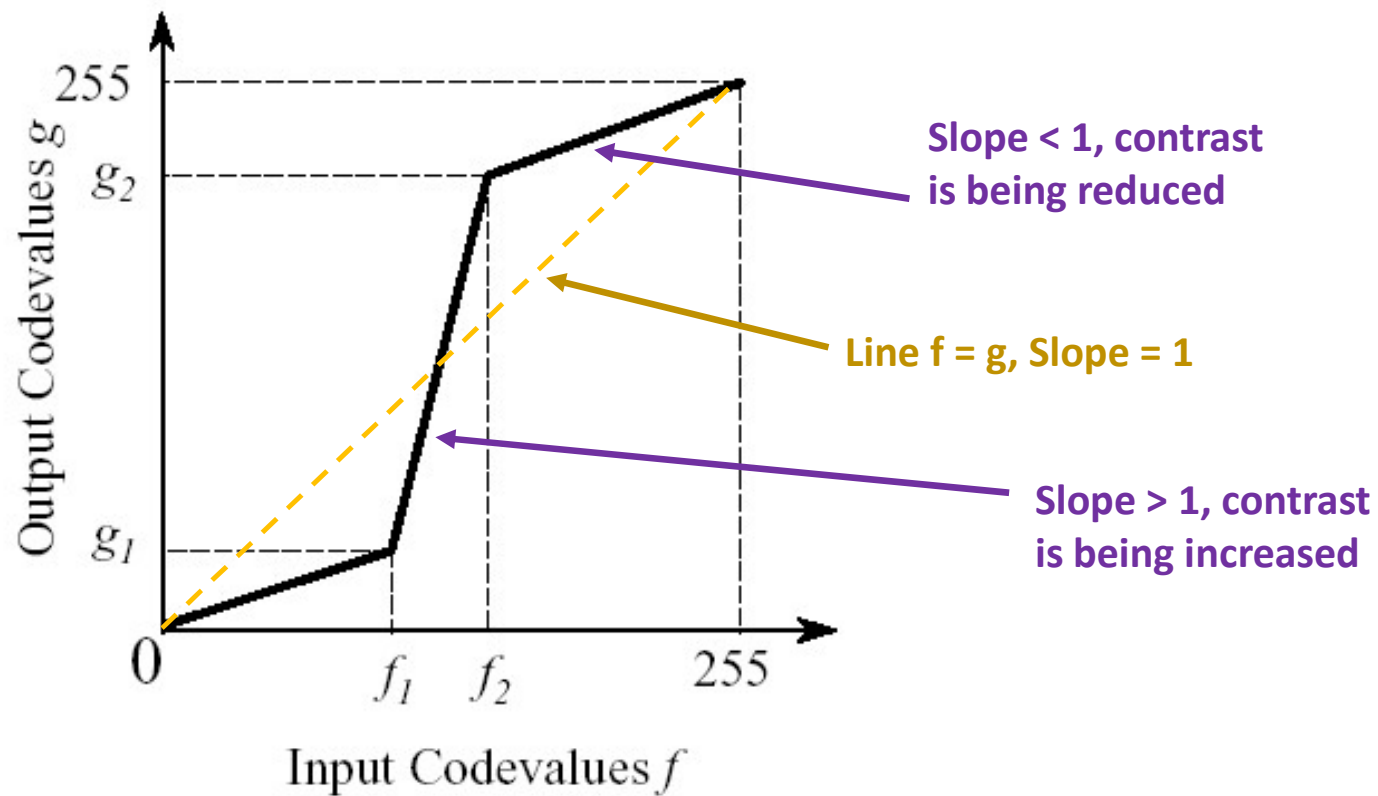
2-8

Can be implemented as a look-up table

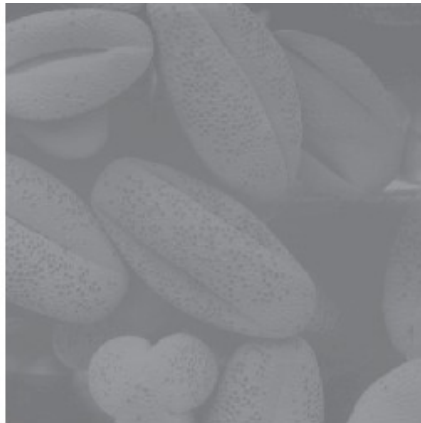


Input f	Output g
0	0
1	0
2	1
3	1
4	2
...	...
...	...
80	40
81	44
82	48
83	52
...	...
...	...
252	254
253	254
254	255
255	255

Contrast Stretching



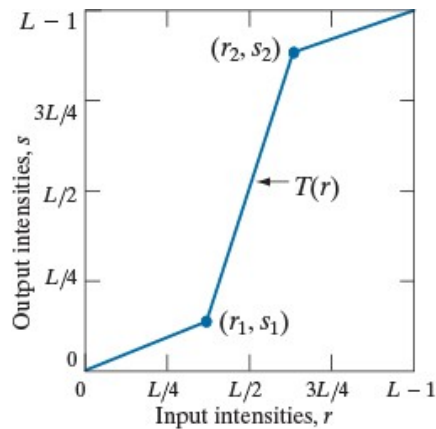
Contrast stretching



Input image

A low-contrast electron microscope image of pollen, magnified 700 times.

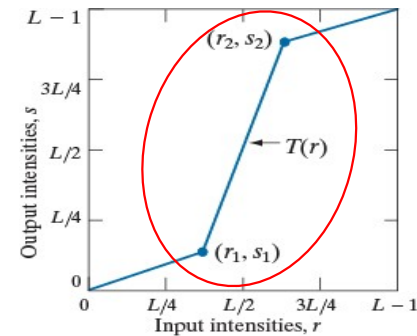
Original image courtesy of Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia



Piecewise linear transformation function



Output image



$$s_1 = ar_1 + b$$

$$s_1 = \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} r_1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} s_1 \\ 1 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ 1 \end{bmatrix}$$

$$s_2 = ar_2 + b$$

$$\begin{bmatrix} s_2 \\ 1 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_2 \\ 1 \end{bmatrix}$$

Solve for a and b

$$\begin{bmatrix} s_1 & s_2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_1 & r_2 \\ 1 & 1 \end{bmatrix}$$

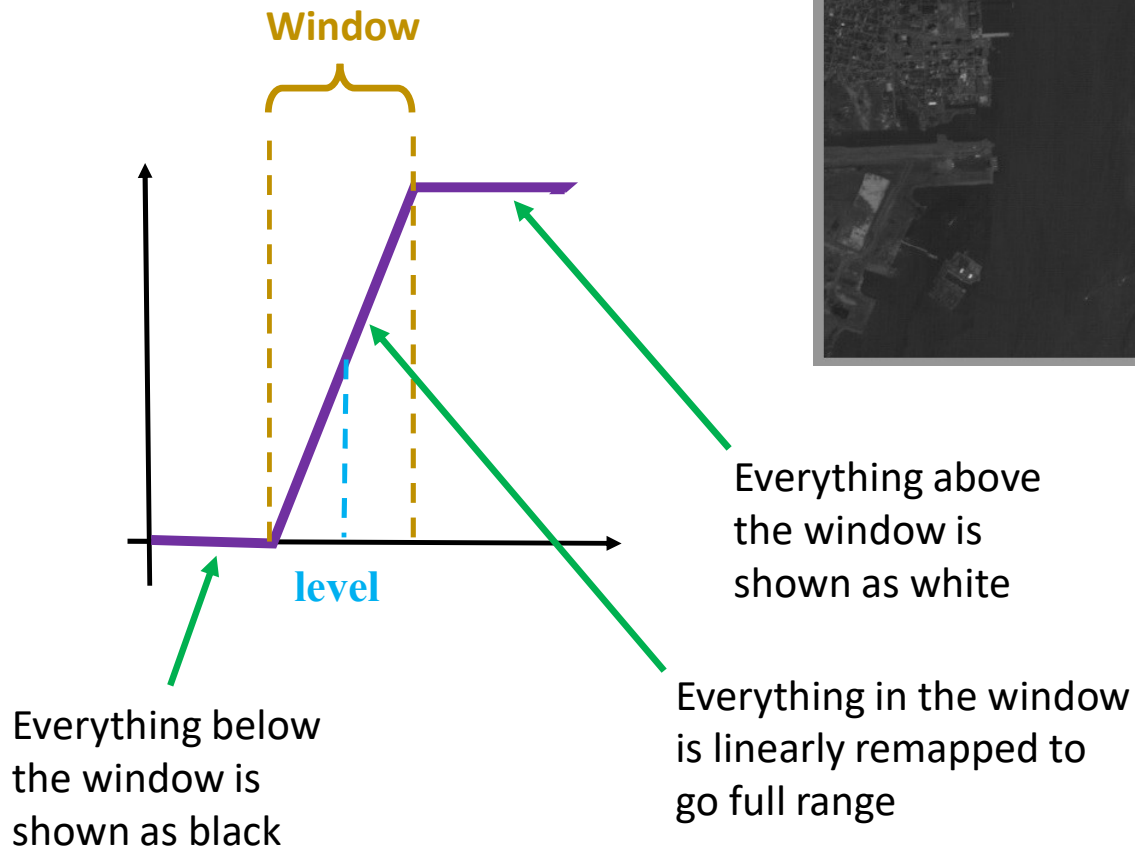
$$\begin{bmatrix} s_1 & s_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} r_1 & r_2 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{s_1 - s_2}{r_1 - r_2} & \frac{r_1 s_2 - r_2 s_1}{r_1 - r_2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$$

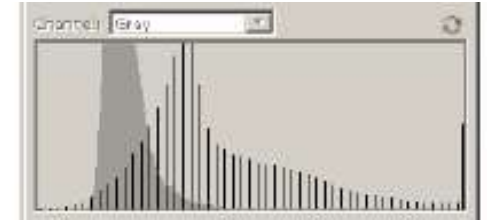
$$a = \frac{s_1 - s_2}{r_1 - r_2}$$

$$b = \frac{r_1 s_2 - r_2 s_1}{r_1 - r_2}$$

Windows and Levels

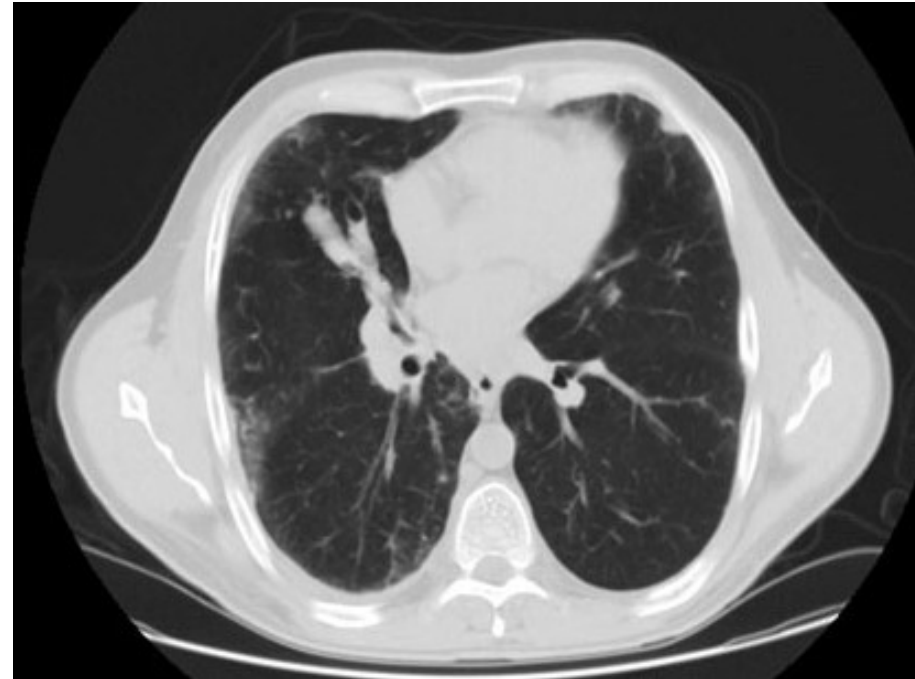
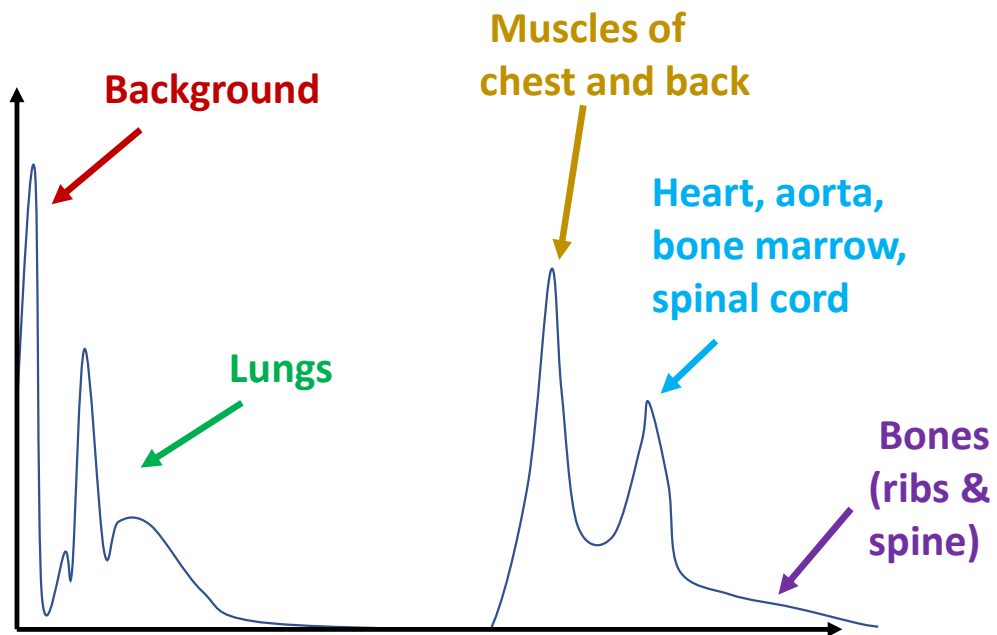


Original 512x512 8-bit

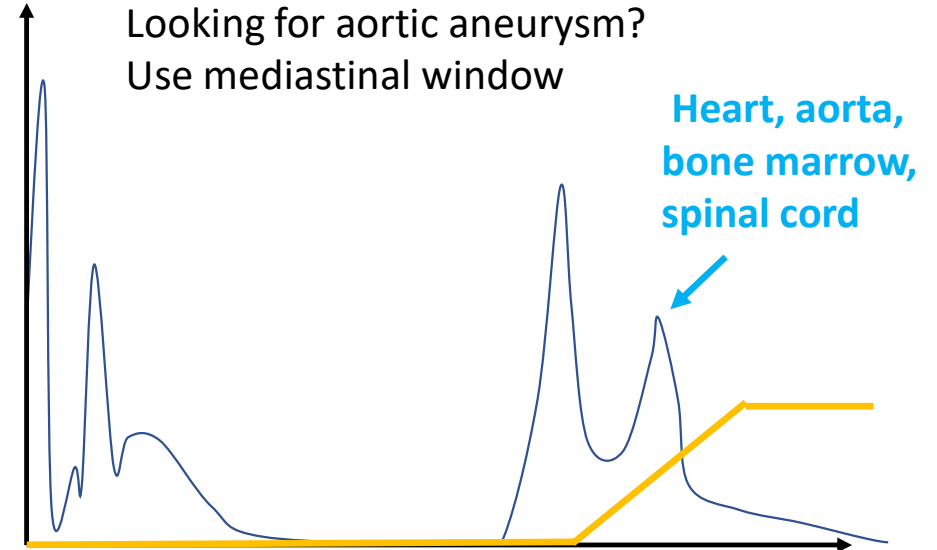
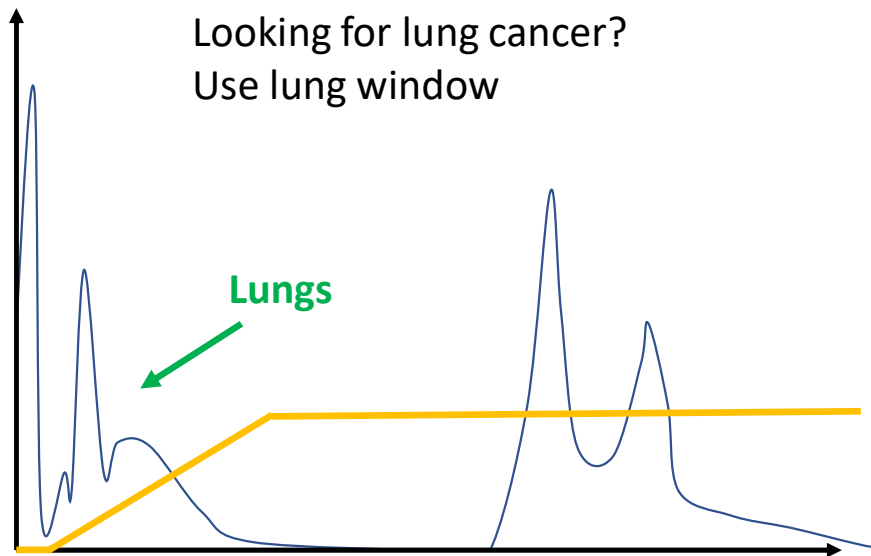


“Windows and Levels” commonly used in medical imaging

Example: CT Chest Scan



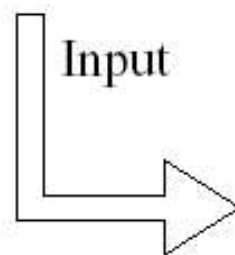
CT Chest Scans: Standard Windows



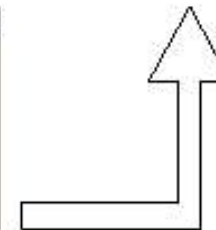
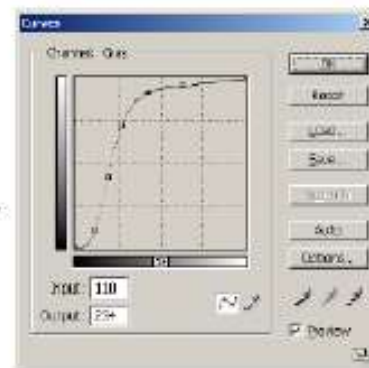
Mediastinum = area between the lungs, includes the heart, major blood vessels, trachea, esophagus, etc.



Example:
Applying a
LUT using
Photoshop
“Curves”



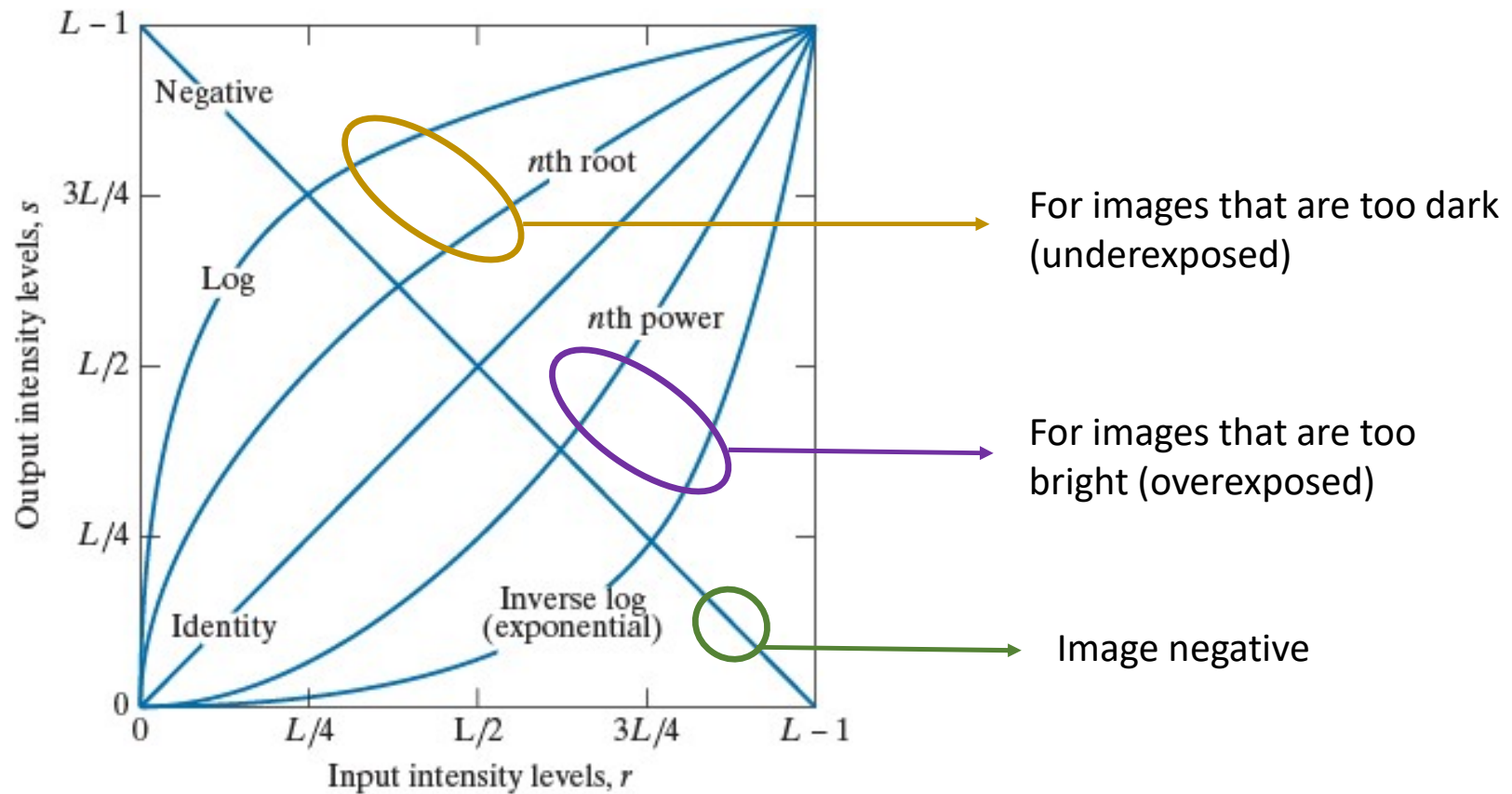
Input



Output

3-7

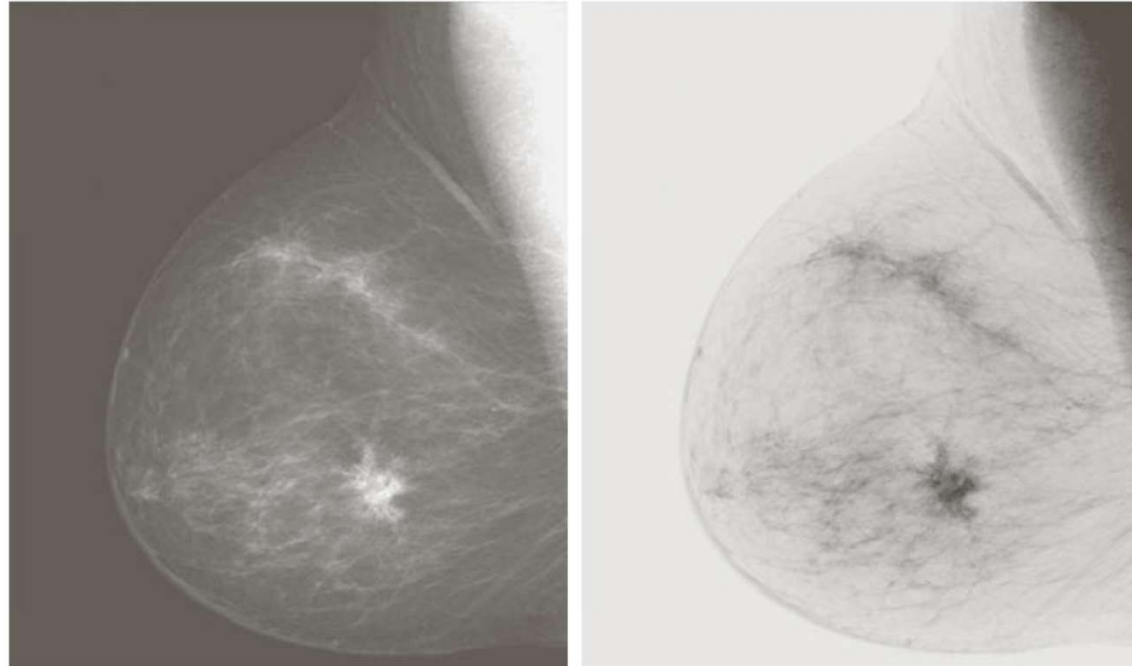
Point/Intensity Transformations



Negative transformation

$$s = (L - 1) - r, \text{ where } r \in [0, L - 1]$$

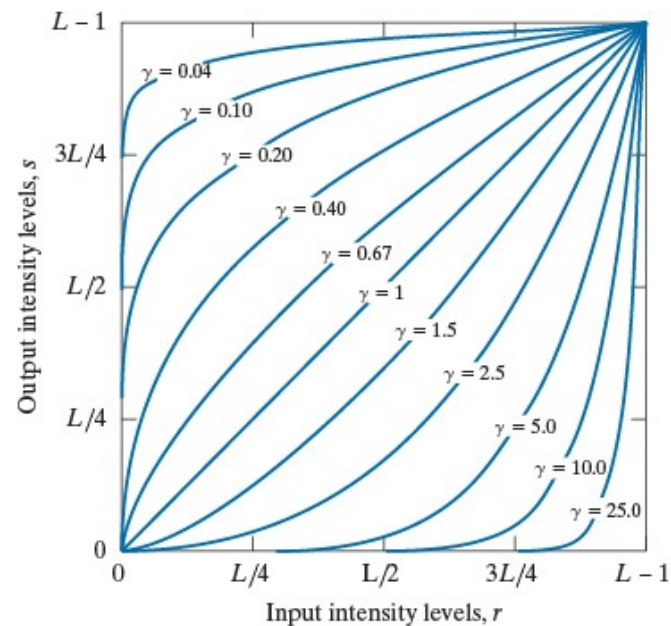
a b
FIGURE 3.4
(a) A digital mammogram.
(b) Negative image obtained using Eq. (3-3).
(Image (a) Courtesy of General Electric Medical Systems.)



Gamma transformation

$$s = (L - 1)c \left(\frac{r}{L - 1} \right)^\gamma, \text{ where } r \in [0, L - 1] \quad c \text{ is often } 1$$

FIGURE 3.6
Plots of the gamma equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.



Gamma transformation

Dark
image

$$\gamma < 1$$

$$s = cr^\gamma$$



a b
c d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation in Eq. (3-5)

with $c = 1$ and $\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Gamma transformation

Light
image

$$\gamma > 1$$

$$s = cr^\gamma$$



a	b
c	d

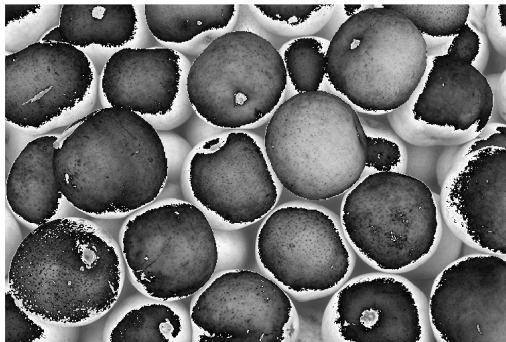
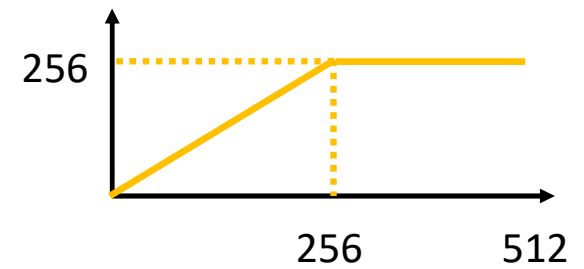
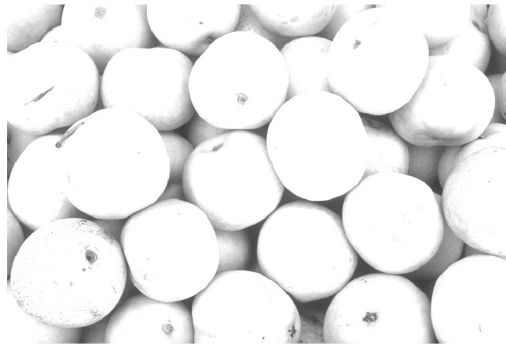
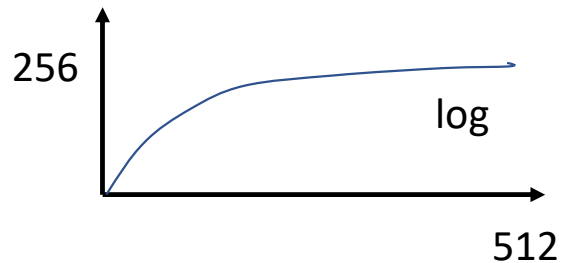
FIGURE 3.9
(a) Aerial image.
(b)–(d) Results
of applying the
transformation
in Eq. (3-5) with
 $\gamma = 3.0, 4.0,$ and
 $5.0,$ respectively.
($c = 1$ in all cases.)
(Original image
courtesy of
NASA.)

Compression of dynamic range

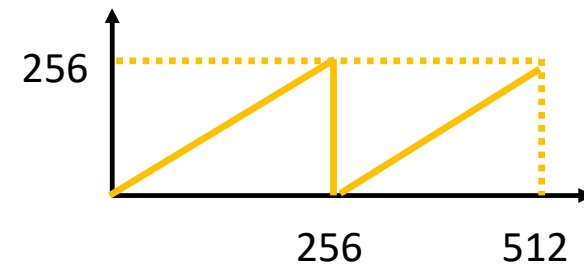
- How to display a 9-bit image on an 8-bit monitor?



8-bit grayscale (0-255) multiply by 2, so artificially get 9-bit grayscale (0-511)



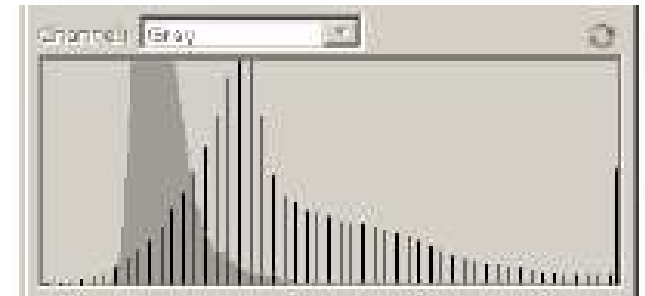
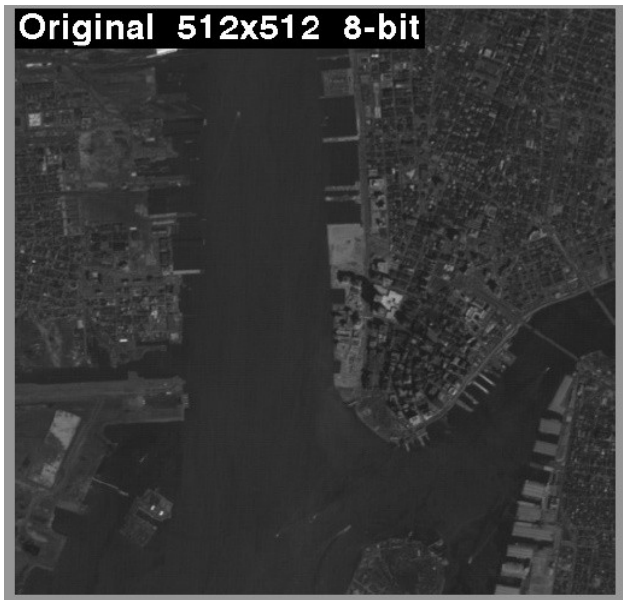
False contours



Outline

- Review of Interpolation & Decimation
- Visual Masking, Weber Law
- Intensity Transformation
- Histogram Equalization

Intuition on Histogram Equalization

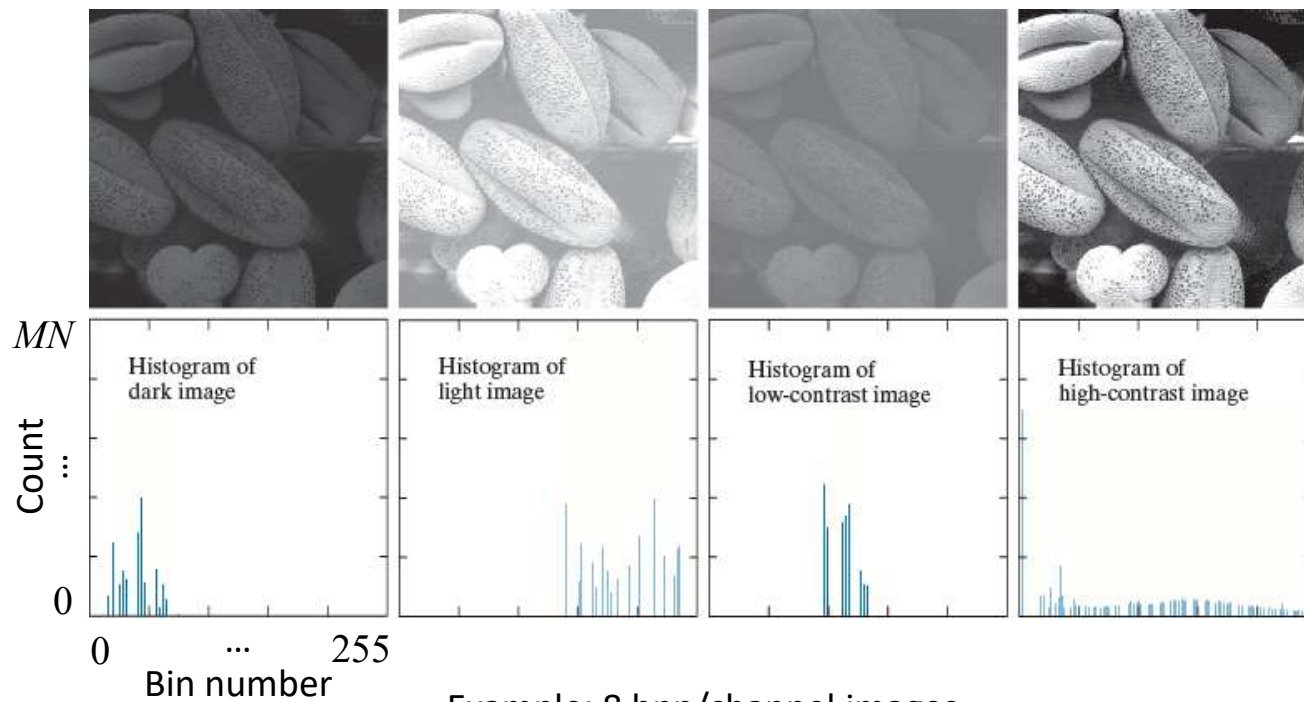


Make full use of the display capabilities of the display device: spread out the histogram

Histogram Examples

a b c d

FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.



Example: 8 bpp/channel images

Normalized histogram

- Probability density function (pdf)
 - Area under curve is 1
- Histogram
 - Area under curve is total number of pixels = MN
- Normalized histogram

$$0 \leq p(x) \leq 1$$

Continuous

$$\text{normalized histogram} = \frac{1}{MN} \text{histogram} \approx \text{pdf} \quad \text{Discrete}$$

$p_f(r)$ is approximate pdf of input image $f(x, y)$

$p_g(s)$ is approximate pdf of output image $g(x, y)$

Cumulative distribution function

- Cumulative distribution function

$$\text{cdf}_f(r) = \int_0^r p_f(w)dw$$

Continuous

$$\text{cdf}_f(r) = \sum_{k=0}^r p_f(k)$$

Discrete

- Monotonically increasing
- Last value is maximum value = 1

$\text{cdf}_f(r)$ is cdf of input image $f(x, y)$

$\text{cdf}_g(s)$ is cdf of output image $g(x, y)$

Histogram equalization

Conditions

- Output intensity values will never be less than corresponding input values

$$s = T(r) \text{ is monotonic increasing in } 0 \leq r \leq L - 1$$

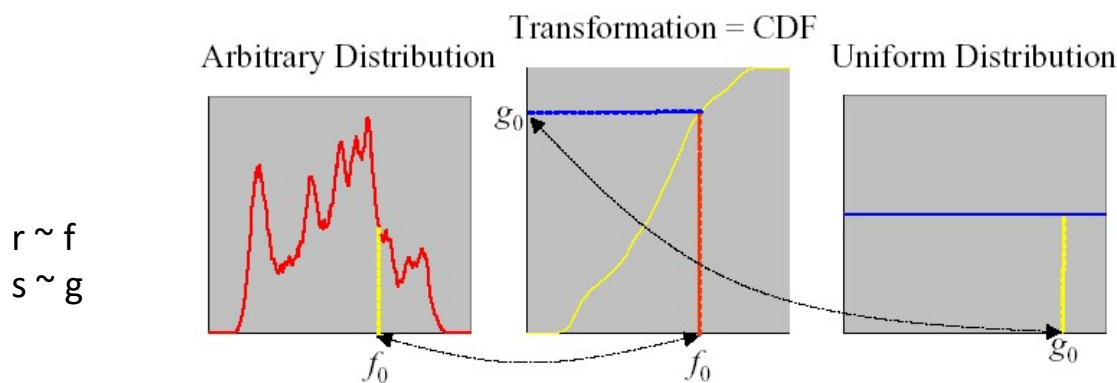
- The range of output intensities is the same as the input

$$0 \leq T(r) \leq L - 1 \quad 0 \leq r \leq L - 1$$

- If we further require the inverse mapping from s back to r

$$r = T^{-1}(s) \quad 0 \leq s \leq L - 1$$

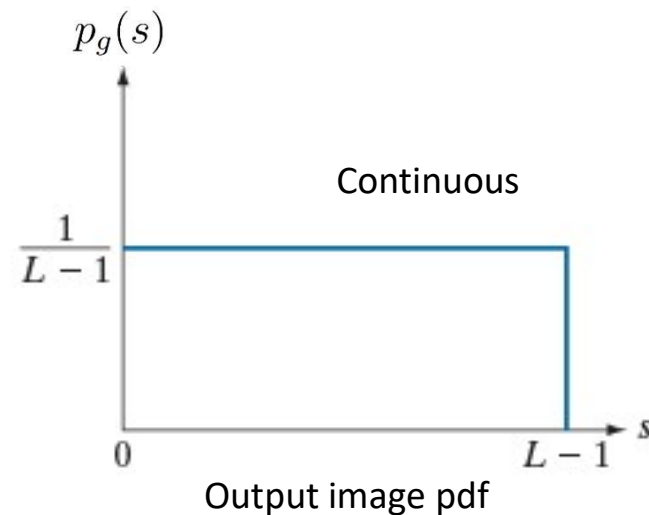
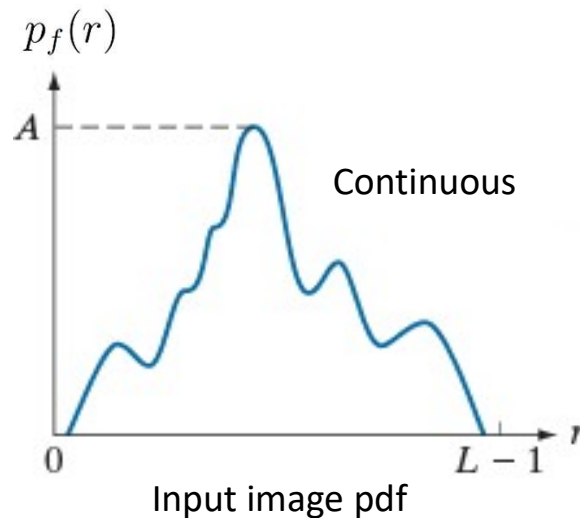
be one-to-one, then $s = T(r)$ is *strictly* monotonic increasing in $0 \leq r \leq L - 1$



The transformation that converts a random variable with a specific distribution into a random variable with a uniform distribution is given by the CDF of that distribution

Histogram Equalization

Objective: After (forward) mapping, the output image pdf is flat



This objective is satisfied by the histogram equalization intensity transformation

$$s = T(r) = (L - 1) \int_0^r p_f(w)dw$$

Histogram equalization

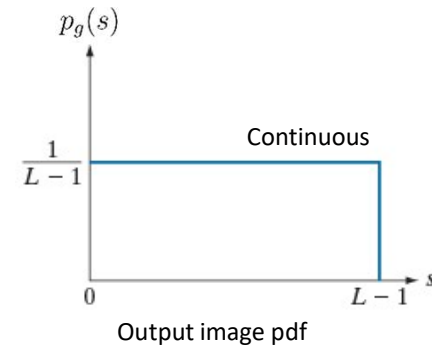
Proof of $s = T(r) = (L - 1) \int_0^r p_f(w)dw$

From calculus, recall the derivative of a definite integral with respect to its upper limit is the integrand evaluated at the limit, i.e.,

$$\begin{aligned}\frac{ds}{dr} &= \frac{dT(r)}{dr} \\ \frac{ds}{dr} &= (L - 1) \frac{d}{dr} \left[\int_0^r p_f(w)dw \right] \\ \frac{ds}{dr} &= (L - 1)p_f(r)\end{aligned}$$

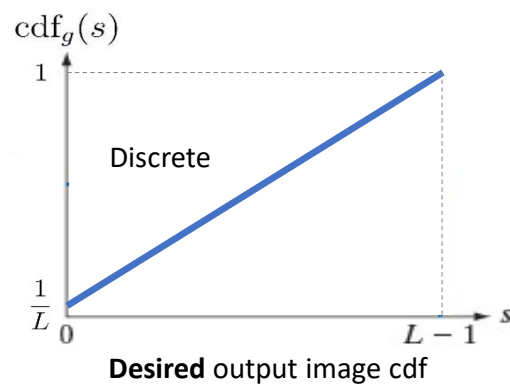
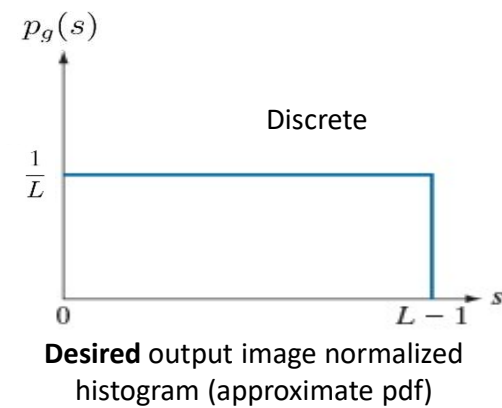
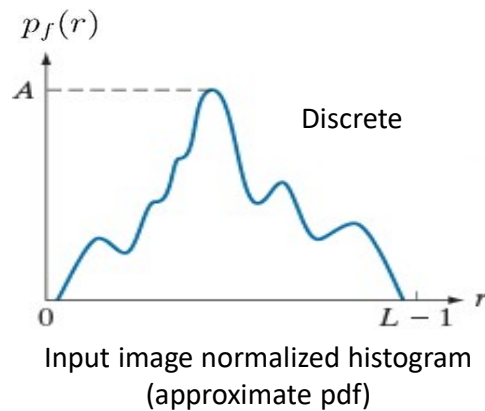
From probability theory, recall if $p_f(r)$ and $T(r)$ are known, and $T(r)$ is continuous and differentiable over the range of values of interest, then the pdf of the transformed (i.e., mapped) variable $s = T(r)$ can be obtained as

$$\begin{aligned}p_g(s) &= p_f(r) \left| \frac{dr}{ds} \right| \\ p_g(s) &= p_f(r) \left| \frac{1}{(L - 1)p_f(r)} \right| \\ p_g(s) &= \frac{1}{L - 1} \quad 0 \leq s \leq L - 1\end{aligned}$$



Histogram Equalization

Objective: for **discrete** images, after (forward) mapping, **desired** output image approximate pdf is flat



$$p_g(s) = \frac{1}{L} \quad s = 0, 1, 2, \dots, L-1$$

$$\text{cdf}_g(s) = \sum_{k=0}^s p_g(k) = \frac{s+1}{L} \quad s = 0, 1, 2, \dots, L-1$$

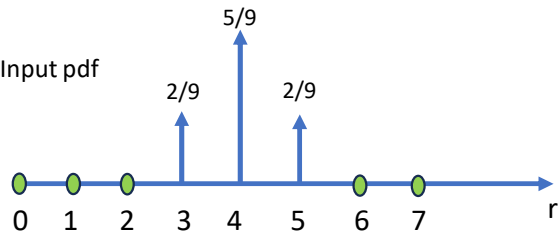
Histogram Equalization

4	4	4	4	4	4
3	4	5	4	3	3
3	5	5	5	3	3
3	4	5	4	4	4
3	5	5	4	4	5
4	4	4	4	4	4

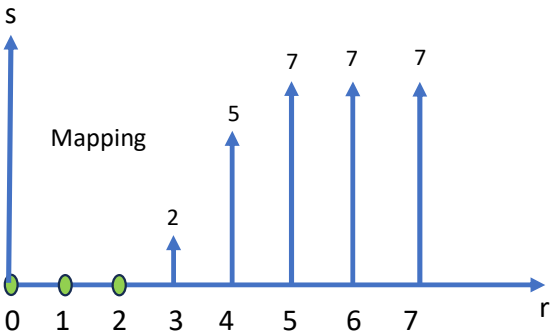
Input image

0	1	2	3	4	5	6	7
0	0	0	8	20	8	0	0

Histogram



Gray Level	# pixels n_k	$r_k = n_k/36$	CDF = s_k	$s_k * 7$	Histogram equalization level
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	8	$2/9$	$2/9$	1.5555	2
4	20	$5/9$	$7/9$	5.4444	5
5	8	$2/9$	1	7	7
6	0	0	1	7	7
7	0	0	1	7	7



5	5	5	5	5	5
2	5	7	5	2	2
2	7	7	7	2	2
2	5	7	5	5	5
2	7	7	5	5	7
5	5	5	5	5	5

Output image

Histogram Equalization

4	4	4	4	4	4
3	4	5	4	3	3
3	5	5	5	3	3
3	4	5	4	4	4
3	5	5	4	4	5
4	4	4	4	4	4

Input image

5	5	5	5	5	5
2	5	7	5	2	2
2	7	7	7	2	2
2	5	7	5	5	5
2	7	7	5	5	7
5	5	5	5	5	5

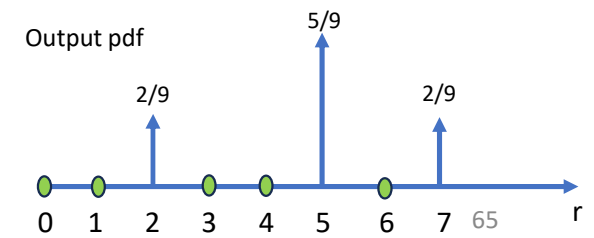
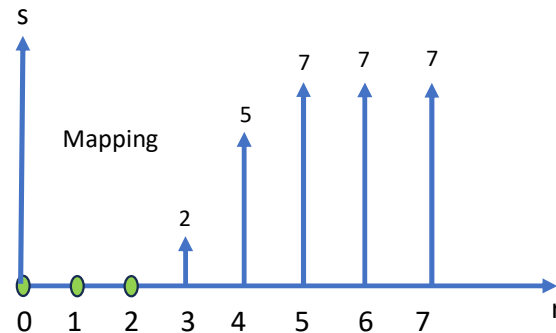
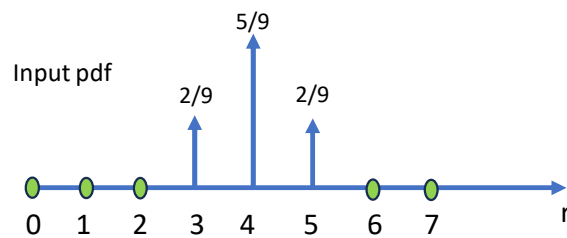
Output image

0	1	2	3	4	5	6	7
0	0	0	8	20	8	0	0

Histogram

0	1	2	3	4	5	6	7
0	0	8	0	0	20	0	8

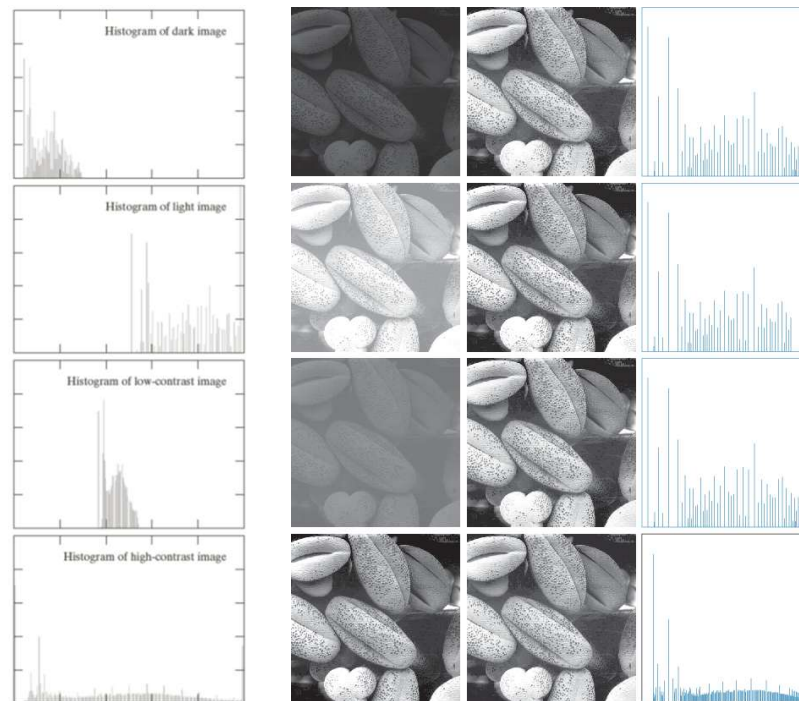
Histogram



Histogram equalization

Discrete

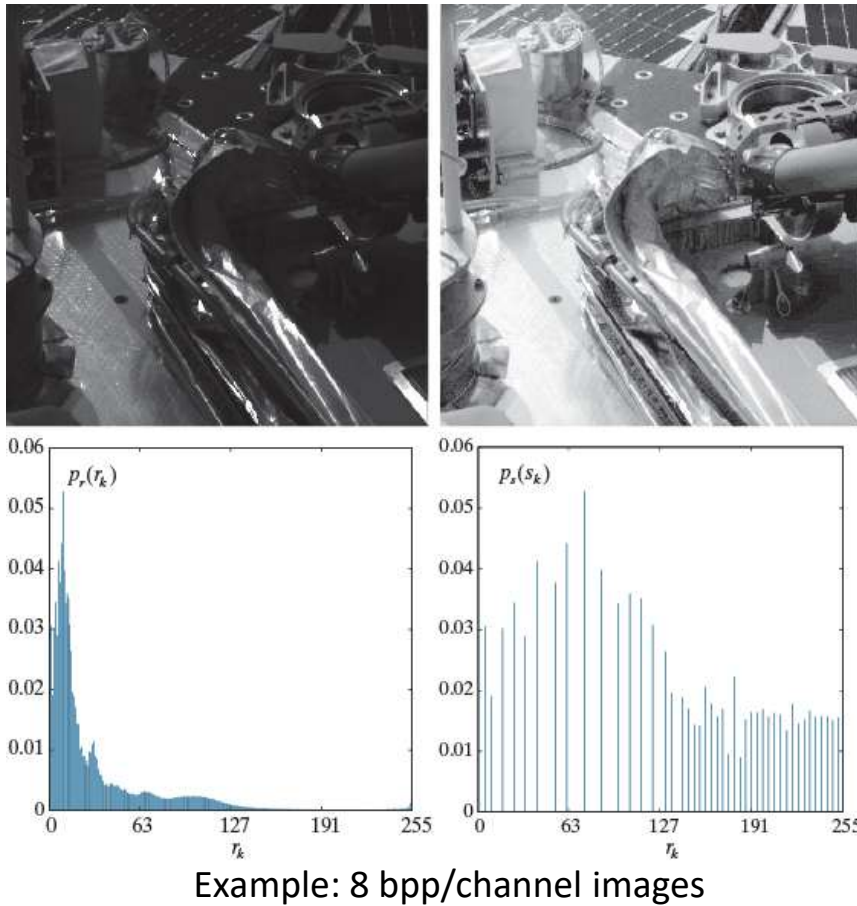
$$s = T(r) = (L - 1)\text{cdf}_f(r) \quad r = 0, 1, 2, \dots, L - 1$$



In general,
discrete pdf
will not be flat

FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).

Histogram equalization



a b
c d

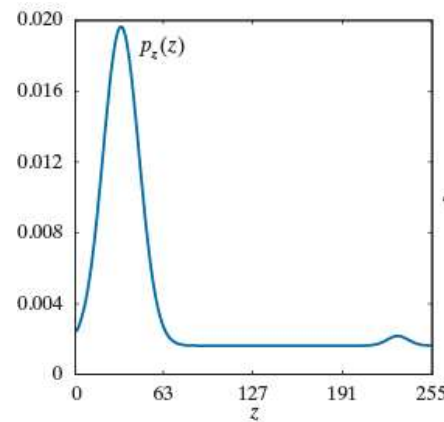
FIGURE 3.22

(a) Image from Phoenix Lander.
(b) Result of histogram equalization.
(c) Histogram of image (a).
(d) Histogram of image (b).
(Original image courtesy of NASA.)

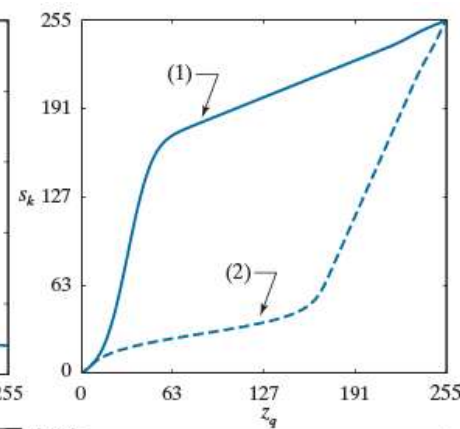
In general,
discrete pdf
will not be flat

Histogram matching

Specified
histogram to
match

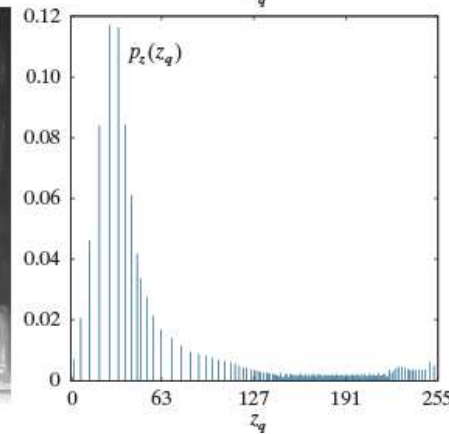


Intensity
transformation



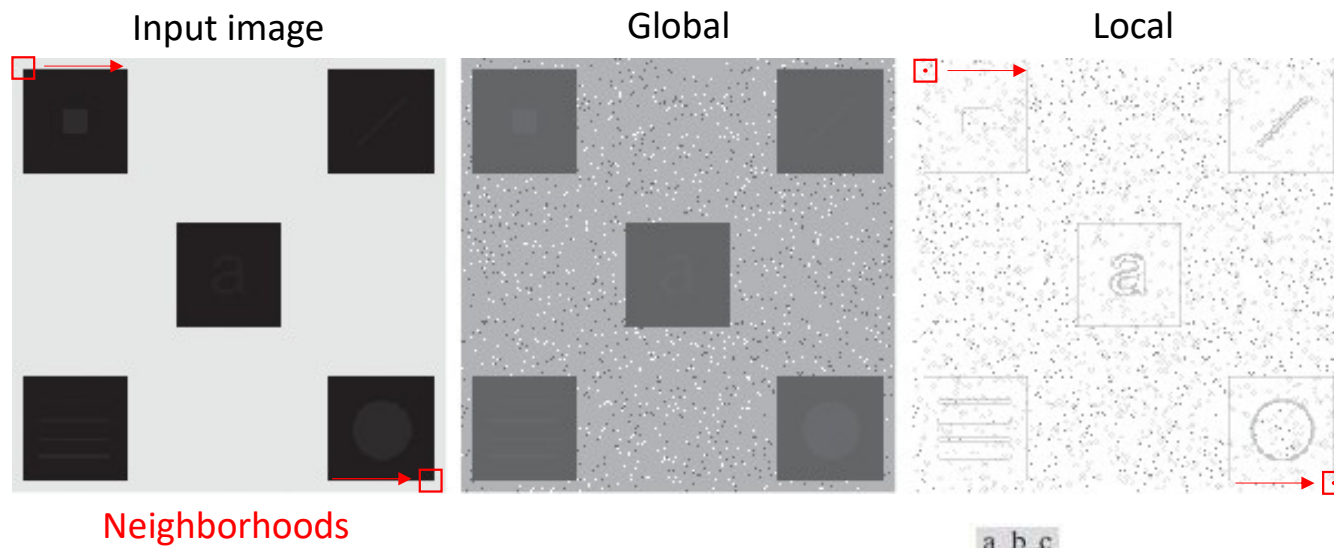
a b
c d

FIGURE 3.26
Histogram
specification.
(a) Specified histo-
gram.
(b) Transformation
 $G(z_q)$, labeled (1),
and $G^{-1}(s_k)$,
labeled (2).
(c) Result of
histogram
specification.
(d) Histogram of
image (c).

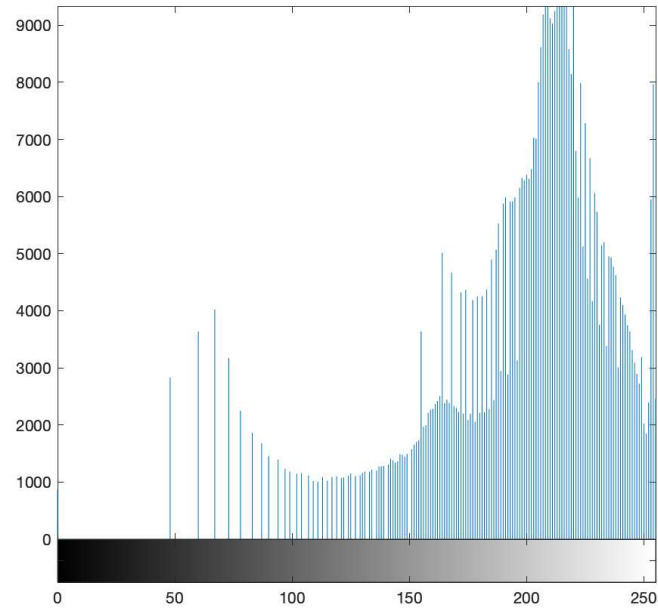


Histogram of
intensity
transformed
image

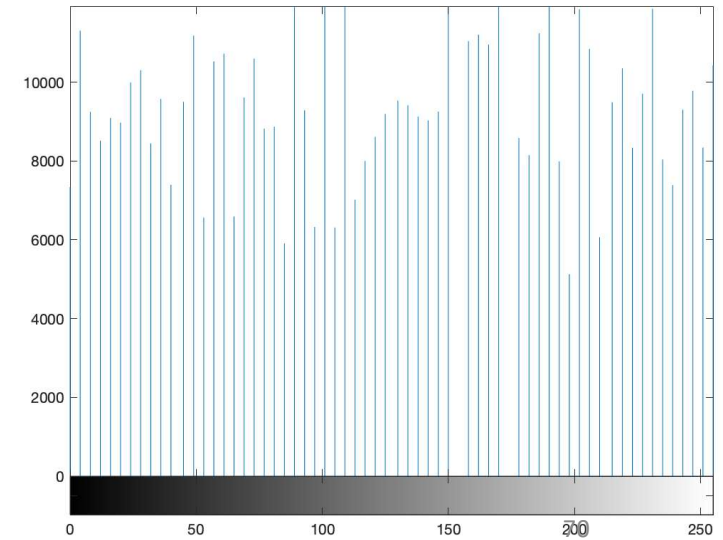
Local histogram equalization



a b c
FIGURE 3.32
(a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.



Histogram is not flat
It is “flat on the average”





Original



"Auto Levels"



Histogram Equalized



"Curves"



"Levels", $\gamma = 1.65$



"Brightness/Contrast"

